THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — MATHEMATICS (Standard) · FULL SYLLABUS PRACTICE PAPER

Session 2026-27 · CBSE / Directorate of Education pattern · Sub code 041

Class: 10	Subject: Mathematics	Set 01	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This Question Paper has 5 Sections 'A', 'B', 'C', 'D' and 'E'.
2. Section A has 20 MCQs carrying 1 mark each (Q19–20 are Assertion–Reason questions).
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based questions (04 marks each) with sub-parts of the values of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 5 marks, 2 questions of 3 marks and 2 questions of 2 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E. You have to attempt only one option out of the given options.
8. Draw neat figures wherever required. Take π = 22/7 wherever required, if not stated.
9. There is no provision of negative marking. Use of calculator is not permitted.
10. Please write down the serial number of the question before attempting it.

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is **1**

(a) 2 (b) 1 (c) 3 (d) 4

2. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is **1**

(a) 10 (b) 5 (c) −5 (d) −10

3. The pair of linear equations $x + 2y - 5 = 0$ and $-3x - 6y + 15 = 0$ has **1**

(a) a unique solution (b) exactly two solutions
(c) infinitely many solutions (d) no solution

4. The values of k for which the equation $2x^2 + kx + 3 = 0$ has equal roots are **1**

(a) $\pm\sqrt{6}$ (b) $\pm 2\sqrt{6}$
(c) ± 24 (d) 6, −6 only if k > 0

5. Which of the following quadratic equations has two distinct real roots? **1**

(a) $x^2 + x + 1 = 0$ (b) $2x^2 - 6x + 3 = 0$
(c) $x^2 - 4x + 4 = 0$ (d) $3x^2 + 2x + 1 = 0$

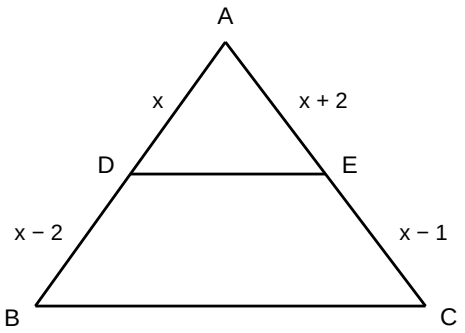
6. The 10th term from the end of the AP 4, 9, 14, …, 254 is **1**

(a) 199 (b) 204 (c) 209 (d) 214

7. The nth term of an AP is $3n + 5$. The sum of its first 10 terms is **1**

(a) 215 (b) 185 (c) 230 (d) 245

8. In the given figure, DE ∥ BC. If AD = x, DB = $x - 2$, AE = $x + 2$ and EC = $x - 1$, then the value of x is **1**



(a) 3 (b) 5 (c) 6 (d) 4

9. ΔABC ~ ΔPQR and ar(ΔABC) : ar(ΔPQR) = 9 : 16. If AB = 6 cm, then PQ equals **1**

(a) 8 cm (b) 4.5 cm (c) 9 cm (d) 10.5 cm

10. The point on the x-axis which is equidistant from (3, 2) and (−5, −2) is **1**

(a) (1, 0) (b) (0, −1) (c) (−1, 0) (d) (−2, 0)

11. If $4 \tan \theta = 3$, then $\frac{4 \sin \theta - \cos \theta}{4 \sin \theta + \cos \theta}$ equals **1**

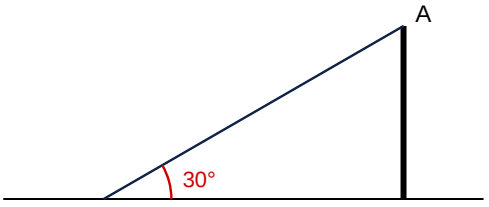
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

12. Which of the following statements is INCORRECT? **1**

(a) $\sin 30° = \cos 60°$ (b) $\sec 60° = 2 \cos 60°$

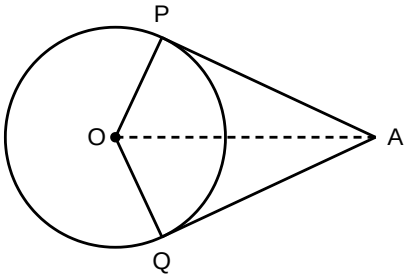
(c) $\tan 45° \cdot \cot 45° = 1$ (d) $\sin^2 45° + \cos^2 45° = 1$

13. The angle of elevation of the top of a tower from a point on the ground, 30 m away from its foot, is 30°. The height of the tower is **1**



(a) $10\sqrt{3}$ m (b) $30\sqrt{3}$ m (c) 15 m (d) 10 m

14. In the figure, AP and AQ are tangents from A to a circle with centre O. If ∠PAQ = 80°, then ∠POA is **1**



(a) 40° (b) 80° (c) 50° (d) 100°

15. The perimeter of a square of side 22 cm is equal to the circumference of a circle. The radius of the circle is **1**

(a) 7 cm (b) 22 cm (c) 28 cm (d) 14 cm

16. A solid sphere of radius 3 cm is melted and recast into small spheres of radius 1 cm. The number of small spheres formed is **1**

(a) 9 (b) 27 (c) 18 (d) 81

17. For the distribution below, the sum of the lower limits of the median class and the modal class is **1**

Class	0–10	10–20	20–30	30–40	40–50
Frequency	6	14	12	18	10

(a) 40 (b) 60 (c) 50 (d) 70

18. Two coins are tossed together. The probability of getting at most one head is **1**

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{3}{4}$

19. **Assertion (A):** The probability that a leap year selected at random has 53 Sundays is 2/7. **1**
Reason (R): A leap year has 52 weeks and 2 extra days; the pair of extra days can occur in 7 equally likely ways, 2 of which contain a Sunday.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The quadratic equation $2x^2 - 4x + 3 = 0$ has two distinct real roots. **1**
Reason (R): A quadratic equation has real roots if and only if its discriminant is non-negative.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

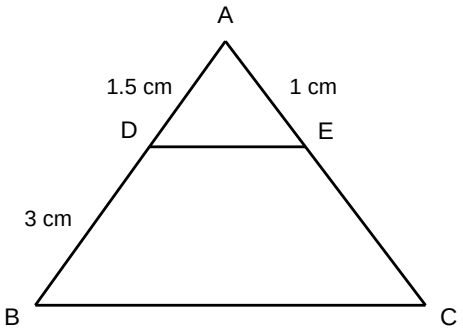
(c) A is true but R is false.

(d) A is false but R is true.

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Find the HCF and LCM of 26 and 91 by the prime factorisation method. Hence verify that HCF × LCM = product of the two numbers. **2**

22. In the figure (DE ∥ BC), AD = 1.5 cm, DB = 3 cm and AE = 1 cm. Find EC. **2**



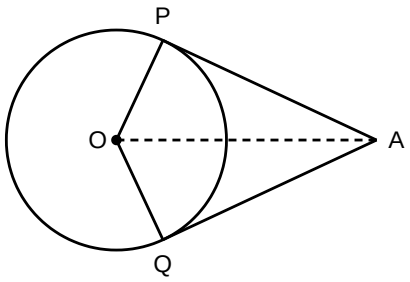
OR

The diagonals of a trapezium ABCD (AB ∥ DC) intersect at O. If AB = 2 CD, find the ratio of the areas of ΔAOB and ΔCOD.

23. Two concentric circles have radii 7 cm and 25 cm. Find the length of the chord of the larger circle which touches the smaller circle. **2**

OR

In the figure, AP and AQ are tangents from A to a circle with centre O and ∠PAQ = 70°. Find ∠OPQ.



24. Find the zeros of the polynomial $6x^2 - 7x - 3$ and verify the relationship between the zeros and the coefficients. **2**

25. A bag contains 5 red, 3 white and 2 black balls. A ball is drawn at random. Find the probability that it is (i) white (ii) neither red nor white. **2**

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Find the ratio in which the *y*-axis divides the line segment joining A(−4, −6) and B(10, 12). Also find the point of division. **3**

27. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. **3**

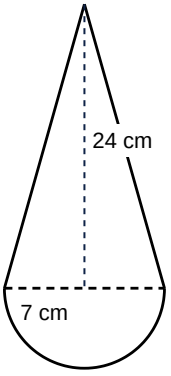
OR

Prove that the lengths of the two tangents drawn from an external point to a circle are equal.

28. The daily wages (in ₹ hundreds) of 35 workers are given. Find the mean wage. **3**

Wage	10–20	20–30	30–40	40–50	50–60
No. of workers	4	7	12	9	3

29. A toy is in the form of a cone of radius 7 cm and height 24 cm mounted on a hemisphere of the same radius. Find the total surface area of the toy. (Use π = 22/7) **3**



30. Prove that $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. **3**

OR

If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, show that $(m^2 - n^2)^2 = 16mn$.

31. A train travels 360 km at a uniform speed. Had the speed been 5 km/h more, it would have taken 1 hour less for the same journey. Find the original speed of the train. **3**

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. Show that the points A(1, 7), B(4, 2), C(−1, −1) and D(−4, 4) are the vertices of a square. **5**

OR

Find the coordinates of the points which divide the line segment joining A(−2, 2) and B(2, 8) into four equal parts.

33. The following table shows the marks of 50 students. Find the median marks. **5**

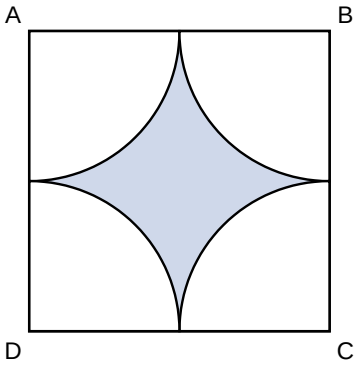
Marks	0–10	10–20	20–30	30–40	40–50	50–60
No. of students	5	8	15	12	6	4

34. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. **5**

OR

In an equilateral triangle ABC, D is a point on side BC such that BD = ⅓ BC. Prove that $9\,AD^2 = 7\,AB^2$.

35. In the figure, ABCD is a square of side 14 cm. With the vertices A, B, C, D as centres, four quadrants of radius 7 cm are drawn (unshaded). Find the area and the perimeter of the shaded region. (Use π = 22/7) **5**



SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. The canteen of Navodaya Public School sells samosas and cups of tea. On Monday, Ritu bought 3 samosas and 2 cups of tea for ₹ 70. On Tuesday, Karan bought 2 samosas and 5 cups of tea for ₹ 76. Let the price of one samosa be ₹ x and of one cup of tea be ₹ y.

(i) Write the pair of linear equations formed by the given information. 1

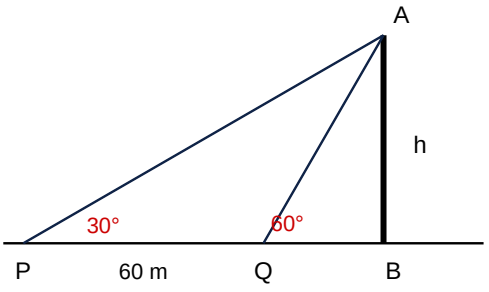
(ii) Without solving, state with a reason whether the pair has a unique solution. 1

(iii) Find the price of one samosa and one cup of tea. 2

OR

(iii) Find the cost of 5 samosas and 3 cups of tea at these prices.

37. The students of Class X measured the height of the school water tank AB using a clinometer. From a point P on the level ground, the angle of elevation of the top A of the tank is 30°. They walk 60 m towards the tank to a point Q, from where the angle of elevation of A is 60°. Let AB = h metres. (Take √3 = 1.73)



(i) Express BQ in terms of h. 1

(ii) Express BP in terms of h. 1

(iii) Find the height of the tank. 2

OR

(iii) Find the distances BQ and BP.

38. Shruti starts a saving plan to buy a bicycle. She saves ₹ 150 in the first month and increases her saving by ₹ 25 every month. Her monthly savings form an AP.

(i) Write the first three terms of the AP and its common difference. 1

(ii) How much does she save in the 12th month? 1

(iii) In which month will she save ₹ 600? 2

OR

(iii) Find her total savings in the first 12 months.